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import numpy as np
import matplotlib.pyplot as plt

# 1. Define the convex curve (e.g., a simple parabola)
x = np.linspace(0.5, 5, 100)
y = 0.5 * x**2 + 1

# 2. Define Point P in the middle of the curve
x_p = 3.0
y_p = 0.5 * x_p**2 + 1

# 3. Calculate the tangent line at Point P
# The derivative of  $y = 0.5x^2 + 1$  is  $dy/dx = x$ 
m = x_p # slope at Point P
# Equation of tangent line:  $y_{tan} = m(x - x_p) + y_p$ 
x_tan = np.linspace(0, 4.5, 100)
y_tan = m * (x_tan - x_p) + y_p

# 4. Calculate the y-intercept (b)
b = m * (0 - x_p) + y_p

# 5. Create the plot
fig, ax = plt.subplots(figsize=(6, 5))

# Plot the convex curve (blue line)
ax.plot(x, y, 'b-', linewidth=2, label='$U = f(S)$')

# Plot the tangent line (dashed line)
ax.plot(x_tan, y_tan, 'k--', linewidth=1.5)

# Plot Point P (black circle)
ax.plot(x_p, y_p, 'ko', markersize=6)
ax.text(x_p + 0.1, y_p - 0.5, 'P', fontsize=12,
        fontweight='bold')

# Plot the y-intercept (b)
ax.plot(0, b, 'ko', markersize=6)
ax.text(-0.3, b, 'b', fontsize=12, fontweight='bold')

# Labeling and formatting
ax.text(3, 10, '$U = f(S)$', color='blue', fontsize=12)
ax.text(1, -2, '$U = T*S + b$', color='black', fontsize=12)

ax.set_xlim(0, 5.5)
ax.set_ylim(-4, 14)
ax.axhline(0, color='black', linewidth=1) # x-axis
ax.axvline(0, color='black', linewidth=1) # y-axis
ax.set_xlabel('S', fontsize=12)

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ax.set_ylabel('U', fontsize=12, rotation=0)

# Remove top and right spines for a cleaner look
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.show()
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